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Thermal Conductivity Reduction in Heterostructure Multilayer Composites by Phonon Density of State Mismatch

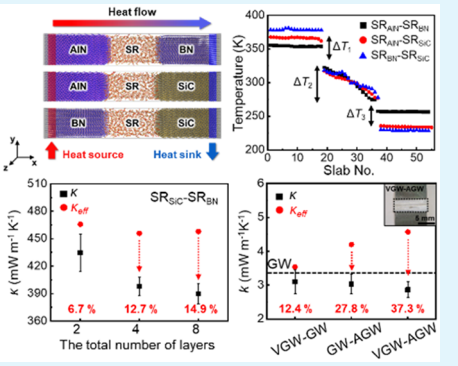
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ABSTRACT: The interfacial phonon transport tuning, depending on the phonon density of states (PDOS) of mating materials, has received considerable attention. However, it has been mainly implemented in nanoscale superlattice structures. Here, we report successful thermal conductivity (κ) modulation in bulk-scale multilayer composites. Single-layer silicone rubber composites, embedded with PDOS-mismatched AlN, BN, or SiC particles, are alternately stacked to construct heterostructure multilayer composites. The molecular dynamics simulation reveals that the SiC−BN combination has the highest PDOS mismatch and interfacial thermal resistance. The experimental analysis also agrees that the SiC−BN combination has the smallest acoustic impedance transmittance and the highest κ reduction ratio of 14.9% (8 layers), demonstrating the reliability of two independent analyses. The κ reduction ratio is as high as 37.3% when PDOS-mismatched vermiculite and silica aerogel particles are alternately impregnated into glass wool, resulting in 2.87 mW m^{-1} K^{-1} in vacuum (24 layers). The PDOS mismatch is a key phonon design factor even for bulk-scale multilayer composites. It might find applications in refrigerators, household appliances, and building insulation panels.



KEYWORDS: *thermal insulation, thermal conductivity reduction, phonon density of state mismatch, heterostructure, multilayer composites*

INTRODUCTION

Thermal conductivity (κ) modulation and efficient thermal management have received considerable attention in household appliances, battery modules, vehicles, and sustainable buildings.[1−4] Diverse fillers, such as boron nitride (BN), MXene, and expanded graphite, have been actively investigated to enhance the κ of polymer−matrix composites.[5−7] On the other hand, insulated wall panels reduce heat transfer to the surrounding environment for the energy-efficient heating and cooling of buildings. Thermal insulation materials in a refrigerator minimize heat transfer and maintain a cold temperature, enabling energy-efficient cooling over a long period of time.[8−11]

The κ of insulation materials has three distinct contributions: solid conduction (κ_{solid}), gas conduction (κ_{gas}), and radiative transfer (κ_{rad}).[12,13] The κ_{rad} is typically negligible at room or lower temperatures.[12,13] Therefore, various strategies have focused on decreasing κ_{solid} and κ_{gas}.[12,13] Porous thermal insulation materials, such as polyurethane (PU) foam, have been widely employed for refrigerators. The porous structure of the PU foam reduces density and interrupts the continuous solid conduction pathway, resulting in a low κ_{solid}.[14,15] It also suppresses κ_{gas} by trapping gas molecules inside the pores.[14−16] It is necessary to decrease the pore size below the mean free path of gas molecules in order to further decrease κ_{gas}.[12,17−19] However, a large decrease in pore size inevitably increases solid skeletons and density, undesirably increasing κ_{solid}.[15,20] The trade-off between κ_{gas} and κ_{solid} limits the minimum κ of PU foam (~22.10 mW m^{-1} K^{-1}).[21] The κ_{gas} can be eliminated by removing gas molecules.[22] Vacuum insulation panels (VIPs) exhibit a significantly lower κ (~1.52 mW m^{-1} K^{-1}).[23] Glass fibers (GFs) have been employed in VIPs to further decrease κ_{solid}.[2,24] The minimal contact area between GFs provides a low κ_{solid}.[23,25] The GFs were also cross-stacked in alternating orientations to further restrict heat transfer between the fibers. Nevertheless, the practical implementation of VIPs has been hindered by the need for airtight encapsulation, gradual loss of vacuum over time, and relatively high production cost.[23]

The κ_{solid} can be further modulated by utilizing boundary thermal resistance between dissimilar materials.[26,27] The phonon transfer or scattering at the interface of dissimilar materials depends on the phonon density of states (PDOS) of each material.[28−32] At the heterogeneous interface, the phonon cannot be transmitted elastically without matching the phonon spectrum frequency of the two materials.[31,32] The mismatched

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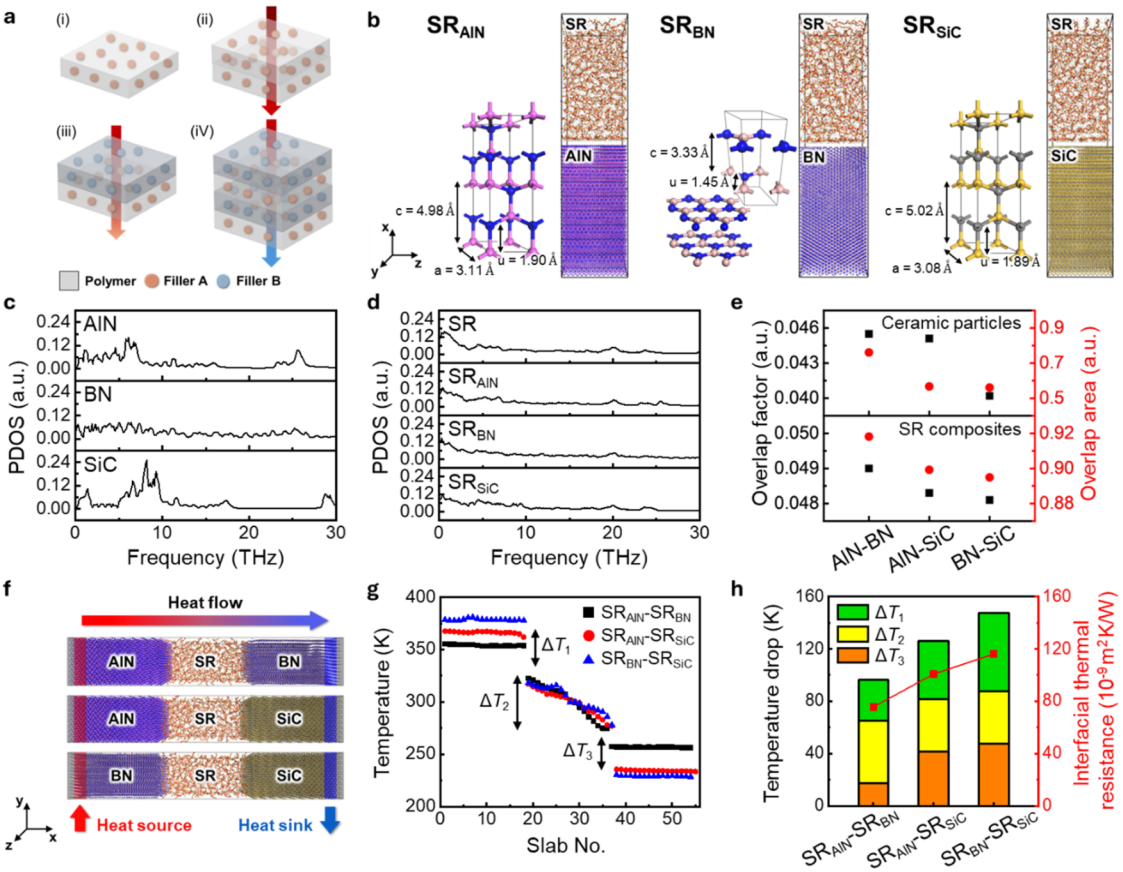


Figure 1. Molecular dynamics simulation: heat transport modulation by phonon density of state mismatch. (a) Schematics of single-layer and multilayer composites. (b) Equilibrated SR-filler models for single-layer composites. The atomic configurations of lattice structures are also provided. (c) PDOS distributions of ceramic (AlN, BN, and SiC) particles. (d) PDOS distributions of pure SR and single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}). (e) Quantification of phonon spectra mismatch using two overlap metrics: overlap factor and overlap area. The metrics for ceramic particles and SR composites are provided at the top and bottom, respectively. A smaller overlap factor (or area) indicates a greater spectral mismatch. (f) NEMD simulation cells to evaluate the interfacial thermal resistance between two layers with dissimilar fillers. The AlN-SR-BN configuration represents the interface between SR_{AlN} and SR_{BN}. Similarly, AlN-SR-SIC and BN-SR-SiC represent SR_{SiC}−SR_{AlN} and SR_{SiC}−SR_{BN} interfaces, respectively. (g) Steady-state temperature profiles along the heat-transport direction (x-axis). The interfacial temperature drop is obtained from slab-to-slab discontinuities. (h) Temperature drop and corresponding interfacial thermal resistance between dissimilar fillers.

phonon spectra would increase the interfacial phonon scattering, decreasing κ_{solid}.[32−34] On the other hand, heat transfer through an ideally smooth interface of dissimilar materials can also be explained by the acoustic mismatch model (AMM) based on the idea that heat transport in solids is carried out by elastic planar waves.[35,36] The energy of the elastic waves is partially transferred through the interface by refraction. It is also partially reflected at the interface, causing a temperature drop. The acoustic impedance (Z) is calculated by $\rho \times v$, where ρ and v are the density and acoustic velocity, respectively.[28,29,37] The normal-direction boundary transmittance (τ) is described by $\tau = 4Z_1Z_2/(Z_1 + Z_2)^2$ for acoustic mismatch in a one-dimensional system.[35]

The phonon spectral mismatch and AMM have been primarily investigated to modulate the interfacial thermal transport of nanoscale superlattice structures where the layer thickness is on the order of a few to several tens of nanometers.[28−32] The high crystallinity of these structures enables the accurate calculation of phonon dispersion for thermal transport modeling.[30] The molecular dynamics (MD) simulations have been employed to model the phonon transport across the interface.[38,39] Despite the successful theoretical development of enhanced phonon scattering at the interface between dissimilar materials, the experimental demonstration has been limited to nanoscale structures. The validity of such interfacial phonon scattering by the PDOS mismatch mechanism has not been sufficiently investigated in bulk-scale multilayer composites, where the individual layer thickness exceeds tens of micrometers.

Here, we report the successful κ modulation in bulk-scale heterostructure multilayer composites by the PDOS mismatch mechanism. Ceramic particles with distinct PDOS, such as aluminum nitride (AlN), BN, and silicon carbide (SiC), are dispersed in the silicone rubber (SR) matrix to synthesize single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}) with a size of $10 \times 10 \times 1.5$ mm^3. The single-layer composites with different fillers are then alternately stacked to construct heterostructure multilayer composites (SR_{AlN}−SR_{BN}, SR_{SiC}−SR_{AlN}, SR_{SiC}−SR_{BN}). The MD simulation reveals that the vibrational spectrum of SR is modulated by the different ceramic fillers with distinct PDOS. The temperature drop and corresponding interfacial thermal resistance increase, at the interface of heterostructure multilayers, as the overlap factor between PDOS spectra decreases. Interestingly, the experimentally measured density and κ do not change even when the same SR_{BN} films are stacked multiple times. In contrast, κ decreases

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as the number of stacking layers increases for heterostructure multilayer composites. The κ reduction ratio, compared with the effective thermal conductivity calculated by the rule of mixtures, increases as the overlap factor decreases. The SR_{SiC}–SR_{BN} composite (stacking number = 8) with the smallest overlap factor (i.e., highest PDOS mismatch) shows the highest κ reduction ratio of 14.9%. The κ reduction ratio also increases as the acoustic impedance transmittance decreases, which is measured by the ultrasonic pulse-echo method. The phonon transport across the interface between layers is still affected by the PDOS mismatch mechanism even for bulk-scale heterostructure multilayer composites. The results from independent analyses (MD, acoustic impedance, and κ measurement) are in close agreement, demonstrating reliability. The PDOS mismatch mechanism can also be expanded to glass wool (GW) for VIP applications. The κ reduction ratio is as high as 37.3% when silica aerogel-impregnated GW (AGW) and vermiculite particle-impregnated GW (VGW) are alternately stacked (total number of layers = 24). The resulting κ is 2.87 mW m^{-1} K^{-1} in a vacuum, which is even smaller than that of GW (3.33 mW m^{-1} K^{-1}).

RESULTS AND DISCUSSION

Molecular Dynamics Simulation: Heat Transport Modulation by Phonon Density of State Mismatch

The phonon transport across the interface of dissimilar materials is affected by the PDOS of each material.[28−32] The interfacial phonon transmission is impeded when the vibrational spectra of the two contacting media are poorly matched. Figure 1a shows the schematics of single-layer and multilayer composites. Ceramic particles with distinct PDOS distributions, such as AlN, BN, and SiC, are dispersed in SR to synthesize a single-layer composite (Figure 1a-i). The single-layer composites with the same or different fillers are then stacked to construct homogeneous or heterostructure multilayer composites (Figures 1a-ii and 1a-iii). The total number of layers is also increased to four and eight by alternately stacking single-layer composites (Figure 1a-iv).

MD simulations have been employed in a wide range of applications, including agricultural and biomedical fields.[40−42] In this study, all-atom MD simulation is carried out to investigate the phonon transport mechanism at the interface of heterostructure multilayer composites. Note that the purpose of the MD simulation is to elucidate the thermal modulation characteristics generated in the polymer matrix domain within the interface region by the presence of the ceramic–polymer interface. In other words, the entire model implemented in the MD environment is located within the interface region, and all analyses are performed focusing on the effective phonon transport characteristics of the matrix region alone due to the presence of ceramic fillers. The construction of a single microstructure and bilayer models is performed using commercial MD simulation software (Materials Studio, Dassault Systèmes). Potential energy minimization of the constructed model and the subsequent thermodynamic stabilization process, as well as heat transfer analysis including PDOS and nonequilibrium molecular dynamics (NEMD) simulations, are performed using an open-source code (LAMMPS, Sandia Lab). All simulations are time-integrated using the velocity-Verlet algorithm with a time step of 0.5 fs. The phonon vibrational characteristics of three fillers (AlN, BN, and SiC) and the modulated vibrational characteristics of the SR influenced by these fillers (i.e., composites) are examined.

The single-layer composite models (SR_{AlN}, SR_{BN}, and SR_{SiC}) considered for the phonon spectrum analysis are shown in Figure 1b. The SR consists of 10 independent chains, each with 150 repeat units (molecular weight of 11211.3 g/mol). The chains are randomly packed into a periodic cell with an initial density of 0.1 g/cm^3 and then densified under the isobaric–isothermal (NPT) ensemble (1 atm and 300 K) until the density converges. Cross-linkers are also employed in the synthesis of SR composites (10 wt %), as will be discussed later, and 43 cross-links are formed at predefined reactive sites to reflect the cross-linking density. Intramolecular and nonbonded interactions in the SR chains are described by a polymer consistent force field (PCFF).[43] The ceramic fillers (AlN,[44,45] BN,[46,47] SiC[48,49]) are modeled using Tersoff potentials parametrized for each lattice,[50] based on the crystal structure shown in Figure 1b (~50 × 50 × 90 $Å^3$). The cross-interaction between the SR and ceramic lattice is represented by the 12–6 Lennard–Jones (LJ) term implemented in the PCFF[51] and universal force field.[52,53] The single-layer and multilayer composite models are assembled by stacking these components along the x-axis. Each model is finally equilibrated under NPT (300 K and 1 atm for 1 ns), and the cell sizes after stabilization are summarized in Supporting Table 1. While the classical force field framework used in this study provides a practical approach for modeling polymer–ceramic hybrid systems, it does not explicitly account for interfacial electronic effects such as polarization and charge redistribution. Modeling phonon transport at interfaces where amorphous polymers and ceramics coexist remains very challenging under conditions that comprehensively consider the electron exchange and chemical reaction. Therefore, the present MD simulation aims solely to understand the characteristics of low-frequency (acoustic) phonon spectrum mismatch and provide qualitative insights into its effects on interfacial heat transport.

The PDOS (D) is obtained by the Fourier transform of the atomic velocity autocorrelation function.[53,54]

$$D(\omega) = \int_0^{\tau} \langle v(0)\cdot v(t)\rangle e^{-2\pi\omega t} dt \quad (1)$$

where ω is the frequency, τ is the total time, $v(t)$ represents the velocity at time t, and bracket $\langle\rangle$ represents the numerical average of the velocity autocorrelation function. The computational details are provided in the Supporting Information. The PDOS distributions of the ceramic fillers and pristine SR are calculated using atomistic models composed solely of each individual material (Supporting Figure 1). The PDOS is obtained by sampling the velocities of all atoms within each model. Figure 1c shows the PDOS distributions of the ceramic fillers. The results show that each inorganic lattice material exhibits a distinct PDOS distribution according to its own vibrational characteristics. Interestingly, such distinct PDOS also affects the SR domain when the SR is adjacent to the filler region. Figure 1d shows the PDOS of pristine SR and modulated SR affected by each filler (i.e., single-layer composites). In these composite systems, the PDOS of the modulated SR is calculated by sampling the velocities of all atoms belonging to the SR matrix within the simulation cell. Even within the same SR matrix, different ceramic fillers modulate the SR vibrational spectrum. This results in different PDOS characteristics of single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}).

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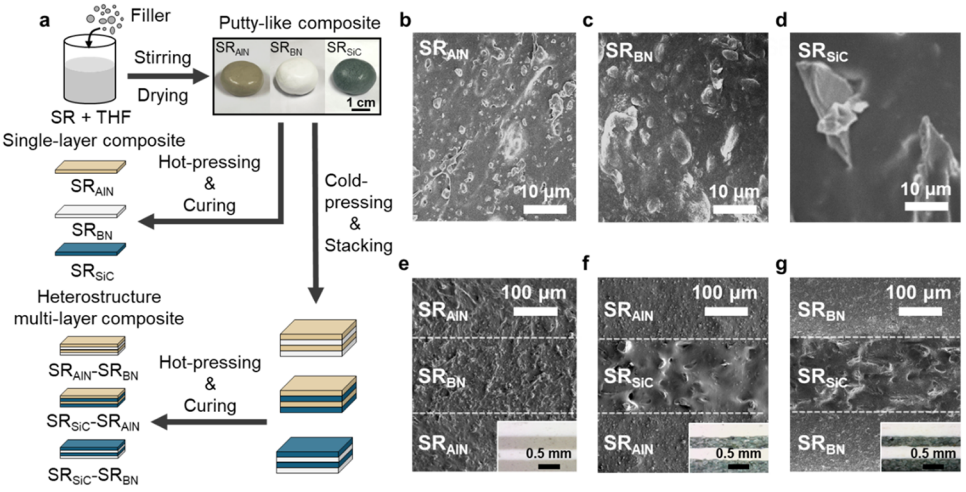


Figure 2. Synthesis of single-layer and multilayer composites. (a) Schematic of the single-layer and multilayer composite synthesis process. (b–d) Cross-sectional SEM images of the single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}). The ceramic particles are uniformly dispersed in the SR matrix. (e–g) Cross-sectional SEM images of multilayer composites (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}). A smooth interface between dissimilar layers is observed without any void. Optical images of the multilayer composites are also provided in the inset. The total number of stacking layers is 4.

Two complementary measures are employed to quantify the vibrational compatibility between the two materials (Figure 1e). The overlap factor (S_1) between two PDOS spectra is calculated by eq 2.[54,55]

$$S_1 = \frac{\int D_1(\omega)D_2(\omega)\mathrm{d}\omega}{\int D_1(\omega)\mathrm{d}\omega \int D_2(\omega)\mathrm{d}\omega} \quad (2)$$

The overlap area (S_2) is simply defined as the fraction of the shared region. The PDOS spectra of pure ceramic particles are first investigated (Figure 1e top). The overlap factor decreases (i.e., the degree of mismatch increases) in the order of AlN-BN, SiC-AlN, and SiC-BN combinations. The two different measures show a similar trend. The properties of such filler combinations also affect SR, and consequently, the degree of mismatch is of the same order in the heterostructure bilayer composites (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}).

Furthermore, the heat transport at the interface of heterostructure bilayer composites is investigated by NEMD simulations. Three types of multilayer stack models are constructed (Figure 1f). The AlN-SR-BN configuration represents the interface between SR_{AlN} and SR_{BN}. Similarly, AlN-SR-SiC and BN-SR-SiC represent SR_{SiC}–SR_{AlN} and SR_{SiC}–SR_{BN} interfaces, respectively. For each model stabilized through the MD simulation, 5 Å slabs at both ends of the simulation box are designated as the fixed region. The two ends of the 5–15 Å slabs adjacent to the fixed region are designated as the heat sink and heat source regions, respectively. In the heat source and heat sink regions, thermal energy is added and removed at a constant rate of 0.004 kcal/mol fs to create a nonequilibrium environment. All atoms are time-integrated for 5 ns under the NVE ensemble, while maintaining the conditions. After time integration for 5 ns, a consistent temperature gradient is observed within the observation area depending on the material composition.

As shown in Figure 1g, the steady-state temperature profile along the heat transfer direction exhibits a linear gradient within the SR region and a clear discontinuity at the interface between the filler and polymer. The interfacial temperature drop ($\Delta T_{total} = \Delta T_1 + \Delta T_2 + \Delta T_3$) and corresponding interfacial thermal resistance between dissimilar fillers (R) are shown in Figure 1h. The magnitudes of ΔT_{total} and R increase in the order of SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}. The interfacial region where different materials come into contact exhibits phonon spectral mismatch. Interestingly, the temperature drop at the interface increases as the phonon spectral mismatch increases. A smaller overlap factor (i.e., greater disparity in vibrational characteristics), arising from the incorporated dissimilar fillers, enhances phonon scattering at the interface of heterostructure bilayer composites. The SR_{SiC}–SR_{BN} shows the highest thermal insulation characteristic among the investigated combinations. The MD simulation suggests that the combination of fillers with different PDOS distributions serves as a key design factor for the interfacial thermal transport modulation of multilayer composites.

Synthesis of Single-Layer and Multilayer Composites

Figure 2a shows a schematic of the single-layer and multilayer composite synthesis process (see the Methods section for details). First, the SR and cross-linker are dissolved in tetrahydrofuran (THF), and ceramic particles with distinct PDOS (AlN, BN, and SiC) are added into the SR solution. The average sizes of AlN, BN, and SiC particles are 10, 1, and 74 μm, respectively. The filler concentration is identical (25 vol %) in single-layer composites, which is calculated using the mass and density of fillers and the polymer (Supporting Table 2). The mixture is then tip-sonicated for 10 min to uniformly disperse fillers in the solution. It is further stirred (40 °C, 500 rpm, 3 h) and dried overnight to evaporate THF and obtain a putty-like composite. It is then hot-pressed in a Teflon mold (5 MPa, 180 °C, 10 min) and further cured on a hot plate (200 °C, 4 h) to synthesize a single-layer composite (thickness = ~500 μm). The thickness is controllable by changing the pressing conditions. Figure 2b–d shows the cross-sectional scanning electron microscopy (SEM) images of SR_{AlN}, SR_{BN}, and SR_{SiC}. Note that the subscript indicates the type of fillers in the single-layer composites. The fillers are uniformly dispersed in the SR matrix. The pores are not observed in the composites, which is favorable to minimize defect-induced phonon scattering.[56] The thermal stability of single-layer

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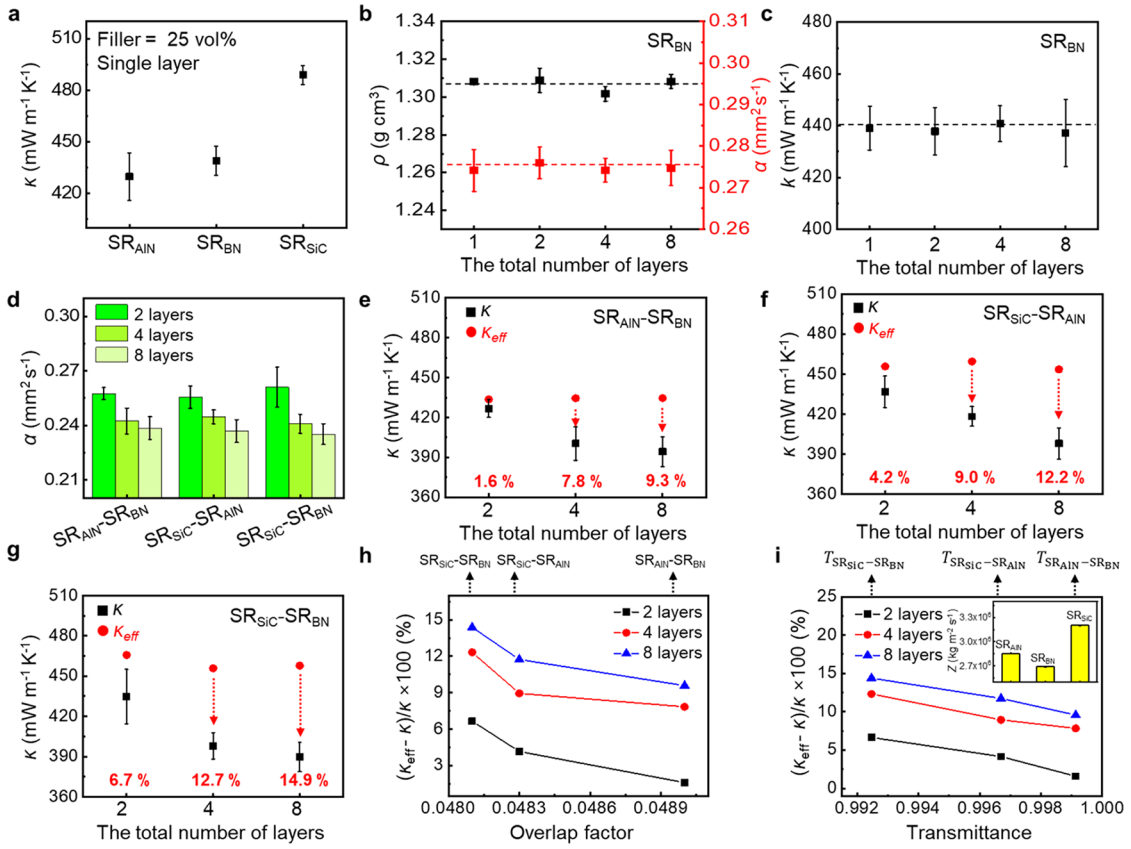


Figure 3. Experimentally measured thermal conductivity of single-layer and multilayer composites. (a) Thermal conductivity of single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}). The error bar represents the standard deviation of the data. (b, c) Density, diffusivity, and thermal conductivity of multilayer SR_{BN} composites are shown as a function of the total number of layers. (d) Diffusivity of heterostructure multilayer composites (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}). (e–g) Thermal conductivity of heterostructure multilayer composites (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}). The effective thermal conductivity is calculated by the rule of mixtures using the thermal conductivities of single-layer composites. The thermal conductivity reduction ratio is noted in red. (h) Experimentally measured κ reduction ratio is compared with the overlap factor calculated by the MD simulation. (i) Experimentally measured κ reduction ratio is shown as a function of the normal-direction boundary transmittance. The acoustic impedance of single-layer composites is provided in the inset.

composites in an air environment is also investigated using thermogravimetric analysis (Supporting Figure 2). The decomposition begins at ~330 °C.[57,58] The residue at 750 °C includes ceramic fillers and SiO_2, which is formed from the oxidation of SR.[58]

In order to synthesize a heterostructure multilayer composite, each putty-like specimen is cold-pressed at 5 MPa to obtain a single-layer composite (thickness = 1 mm). The single-layer composites with different fillers are then alternately stacked to construct heterostructure multilayer composites (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, SR_{SiC}–SR_{BN}). The total number of stacked layers is also changed (2, 4, and 8). Finally, the stacked specimen is hot-pressed in a Teflon mold (5 MPa, 180 °C, 10 min) and further cured on a hot plate (200 °C, 4 h) to obtain a multilayer composite. Figure 2e shows a cross-sectional SEM image of the SR_{AlN}–SR_{BN}. The interface between layers is smooth without any voids or defects. The cross-linking of SR occurs during the hot-pressing and curing process. The excellent layer-to-layer bonding is achieved because the cross-linker constructs covalent bonds between SR even at the interface between layers.[59] An optical image of the multilayer composite is also provided in the inset (total number of stacked layers = 4). The thickness of each layer is decreased from 1 mm to ~300 μm after the hot-pressing process. Nevertheless, the uniform thickness of each layer is well maintained. The excellent bonding interfaces without any voids or defects are also observed for SR_{SiC}–SR_{AlN} and SR_{SiC}–SR_{BN} (Figure 2f,2g). The SEM images without the dashed line are also provided in Supporting Figure 3. As a control, SR_{BN} is stacked multiple times (total number of layers = 2, 4, and 8) to construct multilayer composites with identical fillers. The interface between layers is not distinguishable in the SEM images (Supporting Figure 4). The excellent conformal adhesion between layers minimizes interfacial and defect scatterings.[56]

Experimentally Measured Thermal Conductivity of Single-Layer and Multilayer Composites

Figure 3a shows the κ values of single-layer composites (SR_{AlN}, SR_{BN}, and SR_{SiC}). Note that the κ is experimentally obtained by $\kappa = \alpha\rho C_p$, where α is the thermal diffusivity, ρ is the density, and C_p is the specific heat capacity (see the Methods section for details).[60,61] The α, ρ, and C_p are measured by the laser flash method, Archimedes method, and differential scanning calorimetry, respectively. Supporting Figures 5a and 4c show the ρ, C_p, and α of single-layer composites. Note that the experimentally measured C_p of SR_{AlN} is identical to the theoretical value obtained by the rule of mixtures using the C_p of pure SR and AlN (Supporting Figure 5d). The κ of SR_{AlN} is

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429.9 mW m^{-1} K^{-1}, while SR_{SiC} exhibits the highest κ (488.9 mW m^{-1} K^{-1}).

The effect of the interface between layers on thermal transport is investigated using multilayer SR_{BN} composites. The ρ and α do not change even when the total number of layers is increased from 1 to 8 (Figure 3b). Resultantly, the κ remains almost constant despite the increased number of layers (Figure 3c). This indicates the absence of voids, phonon scattering, and additional resistance at the interface between layers when the same fillers with identical PDOS are dispersed across layers. The effect of chemical bond strength and interphase layer thickness on the interfacial thermal transport should be negligible. The perfect interface without any phonon scattering is enabled by the cross-linker-induced covalent bonding between SR during the hot-pressing and curing process. Note that the C_p of the single-layer SR_{BN} composite (1.22 J g^{-1} K^{-1}) is used for the κ calculation of all the multilayer composites since the composition is identical in all layers. It is also challenging to measure the C_p of large specimens using the small specimen holder of a differential scanning calorimeter.

The thermal transport of heterostructure multilayer composites (SR_{AlN}−SR_{BN}, SR_{SiC}−SR_{AlN}, and SR_{SiC}−SR_{BN}) is also investigated. The change in ρ is negligible as the number of layers is increased (Supporting Figure 6a). This confirms the absence of voids at the interface between the different composite layers. The small random fluctuation in the data is attributed to the different thicknesses of each layer. Note that each composite layer is synthesized by cold pressing, and the multilayer composite is obtained by hot-pressing stacked single layers. There is a small fluctuation in the layer thickness in this manual synthesis process. The C_p of the multilayer composites is calculated by the rule of mixtures using the C_p values of single-layer composites (Supporting Figure 6b). Note that the experimentally measured C_p of SR_{SiC}−SR_{AlN} is identical to the theoretical value obtained by the rule of mixtures using the C_p values of SR_{SiC} and SR_{AlN}, demonstrating the reliability of the rule of mixtures (Supporting Figure 7). In contrast to the multilayer SR_{BN} composites, the α of heterostructure multilayer composites (SR_{AlN}−SR_{BN}, SR_{SiC}−SR_{AlN}, and SR_{SiC}−SR_{BN}) is systematically decreased as the total number of layers is increased (Figure 3d). The different trend in α should arise from the different filler properties, since the cross-linked SR is identical. The effect of interfacial voids or defects is also negligible as discussed earlier. The systematic decrease in α of heterostructure multilayer composites is attributed to the phonon spectral mismatch as discussed in Figure 1.

The decrease in α leads to the decrease in κ of heterostructure multilayer composites. As shown in Figure 3e, κ of SR_{AlN}−SR_{BN} decreases from 426.8 to 394.3 mW m^{-1} K^{-1} as the number of layers increases from 2 to 8. The κ of SR_{AlN}−SR_{BN} is smaller than the κ of single-layer composites (SR_{AlN} = 429.85 mW m^{-1} K^{-1} and SR_{BN} = 439.02 mW m^{-1} K^{-1}). The effective thermal conductivity (κ_{eff}), calculated by the rule of mixtures, is also shown. For example, the κ_{eff} is obtained by eq 3 for two layers.

$$\kappa_{eff} = \frac{L_T}{\left(\frac{L_{AlN}}{\kappa_{AlN}} + \frac{L_{BN}}{\kappa_{BN}}\right)} \quad (3)$$

where L_T is the total thickness of the multilayer composite. The thickness and intrinsic thermal conductivity of each layer are denoted as L_{AlN}, L_{BN}, κ_{AlN}, and κ_{BN}. The subscript indicates the filler type in each layer. The κ_{eff} is obtained by assuming the absence of any phonon scattering at the interface between layers. Note that each layer is constructed manually to synthesize the multilayer composites, and the small random fluctuation in κ_{eff} is due to the slightly different thickness of each layer. The experimentally measured κ of SR_{AlN}−SR_{BN} is smaller than the κ_{eff}. Surprisingly, the reduction ratio, $(\kappa_{eff} - \kappa)/\kappa \times 100\%$, increases from 1.6 to 9.3% as the total number of layers is increased from 2 to 8. This highlights the cumulative effect of interfacial phonon scattering by increasing the number of heterogeneous interfaces in bulk composites. The decreasing trend is very clear even when the number of heterogeneous interfaces is only slightly increased from 2 to 4. The interfacial phonon scattering between layers originates from the inelastic scattering between particulate fillers with mismatched phonon spectra, resulting in partial energy loss and reduced transmission efficiency.[62] The scattering from defects or voids is negligible at the interface between layers, as discussed earlier.

A similar decreasing trend in κ is observed for other heterostructure multilayer composites. The κ of SR_{SiC}−SR_{AlN} decreases from 434.7 to 389.7 mW m^{-1} K^{-1} as the stacking layer number increases from 2 to 8 (Figure 3f). The corresponding κ reduction ratio increases from 4.2 to 12.2%. The κ of SR_{SiC}−SR_{BN} decreases from 434.7 to 389.7 mW m^{-1} K^{-1} with an increasing total layer number (Figure 3g). The corresponding κ reduction ratio increases from 6.7 to 14.9%. The SR_{SiC}−SR_{BN} provides the highest κ reduction ratio (14.9% at layer number = 8) compared with other heterostructure multilayer composites.

The experimentally measured κ reduction ratio is shown as a function of the overlap factor calculated by the MD simulation (Figure 3h). Surprisingly, the experimentally measured reduction trend in κ is perfectly correlated to the MD simulation results. The κ reduction ratio increases as the overlap factor decreases. A smaller overlap factor indicates a greater mismatch in PDOS. The SR_{SiC}−SR_{BN}, with the lowest overlap factor between fillers (0.0481), provides the highest κ reduction ratio. This further supports that the reduction in the κ of heterostructure multilayer composites is attributed to the inelastic scattering between fillers with different PDOSs.

The interfacial thermal resistance ($R_{interface}$) analysis is also carried out. The thermal circuit of the two-layer composite is shown in Supporting Figure 8a. The $R_{interface}$ is extracted by using the thickness and κ of the entire sample and each layer. The $R_{interface}$ decreases as the overlap factor increases (Supporting Figure 8b). The sum of $R_{interface}$ values at all of the interfaces increases as the total number of layers increases. The phonon transport across multiple interfaces, corresponding progressive modulation in $R_{interface}$ at each interface, and length-dependent phonon scattering need to be investigated further in the future.

Although the trend in κ reduction of heterostructure multilayer composites is well matched with the PDOS analysis, the MD simulation provides only localized information at the nanoscale interfacial region. It is difficult to experimentally measure the PDOS of entire bulk-scale composites. On the other hand, the heat transfer through an ideally smooth interface between dissimilar bulk materials can also be explored by the acoustic mismatch model (AMM).[35,36] The heat transfer in solids is carried out by elastic planar waves in the AMM. The energy of elastic waves is partially reflected at the interface, causing a temperature jump, and is partially

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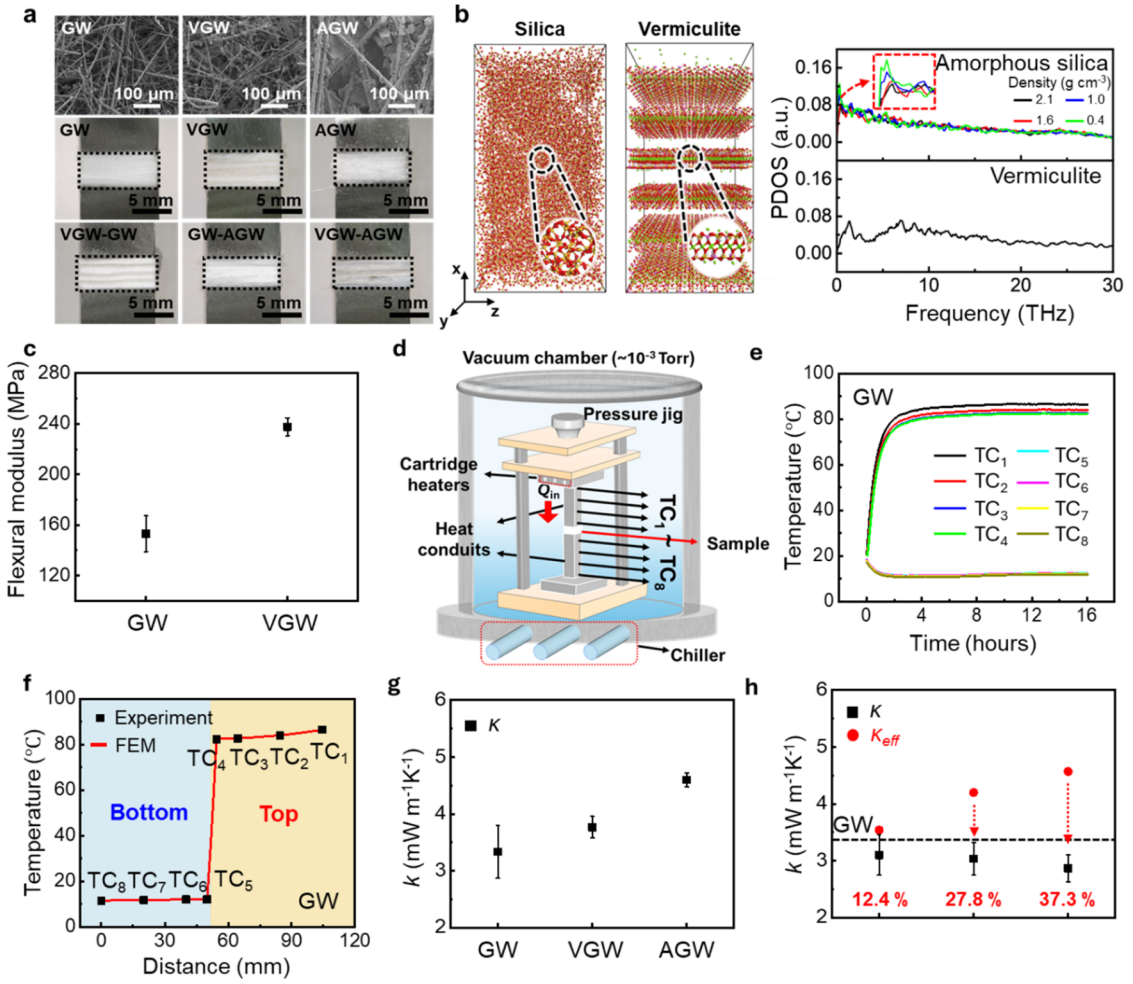


Figure 4. Thermal conductivity reduction of glass wool by phonon density of state mismatch in vacuum. (a) SEM and optical images of GW, VGW, and AGW. Optical images of VGW−GW, GW−AGW, and VGW−AGW sandwiched between heat conduits are also shown. (b) PDOS distributions of amorphous silica and vermiculite calculated from all-atom MD simulations. (c) Flexural modulus of GW and VGW. The total number of layers is identical (32) in both GW and VGW. (d) Schematic of the thermal conductivity measurement setup. (e) Temperatures of stainless-steel heat conduits are shown as a function of time (specimen = GW). (f) Comparison of the experimentally measured temperature profiles and FEM simulation result (specimen = GW). (g) Thermal conductivity of GW, VGW, and AGW. (h) Thermal conductivity of heterostructure multilayer specimens (VGW−GW, GW−AGW, and VGW−AGW). The effective thermal conductivity is calculated by the rule of mixtures using the thermal conductivities of single component specimens. The thermal conductivity reduction ratio is noted in red.

transferred through the interface by refraction. The acoustic impedance (Z) of the single-layer composite is experimentally obtained by $\rho \times v$ (Figure 3i inset). The v comprises longitudinal and transverse components. It is difficult to accurately measure the transverse component due to the strong attenuation in viscoelastic materials.[63] Therefore, only the longitudinal component is typically evaluated in polymer-based composites.[64,65] The Z analysis in this study is also carried out using the longitudinal component experimentally measured by the ultrasonic pulse-echo method (see the Methods section for details).[28,29,37] SR_{BN} has the lowest Z (2.69×10^6 kg m^{-3} s^{-1}), while the Z of SR_{SiC} is the highest (3.20×10^6 kg m^{-3} s^{-1}).

The heterostructure multilayer composite can be considered a quasi-one-dimensional system since heat flows normal to the surface in this study. In such a case, the normal-direction boundary transmittance (τ) is described by $\tau = 4Z_1Z_2/(Z_1 + Z_2)^2$.[35,36] The subscripts indicate each layer. As shown in Figure 3i, the experimentally measured κ reduction ratio decreases as τ increases. The combination of SR_{AlN} and SR_{BN} has the highest τ (0.9991), and the corresponding κ reduction ratio is the lowest (e.g., 1.59% at 2 layers). The lowest τ (0.9925) is achieved by SR_{SiC}−SR_{BN}, resulting in the highest κ reduction ratio (e.g., 14.9% at 8 layers). The experimentally measured Z analysis of bulk composites is closely correlated with the PDOS mismatch analysis of the localized interfacial region by MD simulation. This confirms the importance of employing fillers with different PDOS spectra or Z in order to control the heat flow through heterostructure multilayer composites. The PDOS or Z mismatch-induced scattering at the interface between heterostructure layers is powerful, even when the number of interfaces is small (1−4) in bulk-scale composites.

The consistency between the PDOS analysis and experimentally measured acoustic impedance results supports that the proposed modeling approach captures the essential physics of phonon mismatch-induced interfacial thermal resistance in multilayer composites. The PDOS modulation is not strictly confined to the immediate interfacial region but can extend over larger length scales due to the collective and nonlocal nature of vibrational energy transport in polymer systems.

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Nevertheless, the scale discrepancy between the MD simulation and bulk-scale composites imposes limitations on theoretical analysis. A larger scale simulation needs to be carried out in the future.

Thermal Conductivity Reduction of Glass Wool by Phonon Density of State Mismatch in Vacuum

The κ can be further reduced by eliminating κ_{gas} in vacuum.[22] A significantly lower κ can be obtained using VIPs for refrigerator and building insulation applications.[23] Glass wool (GW) has been intensively investigated for VIPs in the industry since the minimal contact area between fibers decreases κ_{solid}.[2,24] Figure 4a clearly shows the fiber structures of the GW. A further reduction in the κ of GW is explored by impregnating it with dissimilar particles. The SEM images of vermiculite-impregnated GW (VGW) and silica aerogel-impregnated GW (AGW) are shown in Figure 4a. Vermiculite is composed of a group of hydrated, laminar magnesium−aluminum−iron silicate minerals. The chemical formula varies, reflecting its variable composition, but a common representation is $(Mg,Fe,Al)_3(Al,Si)_4O_{10}(OH)_2{\cdot}4H_2O$. Magnesium, iron, and aluminum can substitute for each other in the octahedral layers, and aluminum and silicon are found in the tetrahedral layers, with water molecules (H_2O) present in the interlayer spaces. A detailed impregnation procedure is provided in the Methods section.

Figure 4b compares the PDOS spectra of the silica aerogel and vermiculite particles obtained by all-atom MD simulations. Since there are variations in the composition and structure of the silica aerogel and vermiculite particles, representative conditions are selected for the PDOS calculation. The amorphous silica (SiO_2) phase is generated starting from the β-cristobalite configuration under the Vashishta potential,[66] followed by high-temperature annealing and rapid quenching.[67] By adjustment of the unit cell size and equilibration at 300 K, four amorphous silica models with densities of 2.1, 1.6, 1.0, and 0.4 g cm^{-3} are obtained. It is interesting to note that the PDOS of silica changes as a function of density, which might represent the difference between GW and aerogel. The vermiculite is modeled using CLAYFF[68−70] based on the Mg-vermiculite structure reported by Shirozu et al. ($Mg_3Al_1Si_3O_{10}(OH)_2$).[71] The PDOS is calculated for each equilibrated structure using the Fourier transform of atomic velocity autocorrelation function (eq 1).[53,54]

The PDOS of low-density silica (i.e., silica aerogel) differs from that of high-density silica (GW), suggesting that AGW has a spectral mismatch at the interface between the aerogel and GW. The PDOS mismatch is most pronounced for vermiculite and low-density silica, suggesting that phonon transport is most effectively hindered at the interface where these two particles come into contact. The larger mismatch in PDOS is expected to further decrease κ_{solid}, which is particularly pronounced under vacuum where κ_{gas} is absent. The combination of VGW-AGW is experimentally found to provide the minimum κ value, as will be discussed shortly.

The particle impregnation into GW originally received attention in the industry to increase the mechanical strength. The thermal insulation panel should also have enough mechanical strength for use in household appliances. Figure 4c compares the flexural modulus of GW and VGW, measured using a universal test machine according to the ASTM-C203−99 standard.[72] With vermiculite particle impregnation, the flexural modulus increases from 153 to 238 MPa, corresponding to an enhancement of 55.3%.

The heterostructure multilayer specimens are fabricated by alternately stacking GW, VGW, and AGW. Figure 4a shows the optical images of VGW−GW, GW−AGW, and VGW−AGW. VGW−GW has a total of 24 layers where three layers of VGW and three layers of GW are alternately stacked four times. GW-AGW and VGW-AGW also have a total of 24 layers by alternately stacking three layers of each component. The heterostructure multilayer specimen is then sandwiched between two stainless-steel heat conduits for the κ measurement (Figure 4d). It is difficult to measure the κ of GW by the laser flash method due to the high porosity and ultralow conductivity. Therefore, a laboratory-built κ measurement setup is constructed according to the ASTM-D5470 standard (see the Methods section for details).[73,74] The top conduit is equipped with three cartridge heaters, and the bottom conduit is connected to a chiller placed beneath the vacuum chamber. The chiller maintains a low temperature of the bottom stainless-steel conduit to promote heat transfer through the GW specimen with a low κ. The temperature distribution is measured by using eight T-type thermocouples embedded in stainless-steel heat conduits (TC_1−TC_8). The specimen thickness and applied pressure (0.15−0.2 MPa) are monitored using thickness and pressure sensors, respectively. The chamber is evacuated by using a vacuum pump to eliminate κ_{gas}. Supporting Figure 9 shows the chamber pressure ($\sim 10^{-3}$ Torr) as a function of pumping time. The top heat conduit is simultaneously heated, while vacuuming the chamber. Figure 4e shows the temperature profiles of stainless-steel heat conduits as a function of time (spectra = GW). The temperatures of the top conduit (TC_1−TC_4) increase as the heat flux is applied to the cartridge heaters. In contrast, the temperatures of the bottom conduit (TC_5−TC_8) decrease as the chamber is vacuumed, indicating the elimination of convection heat transfer. The chiller is also operative during the vacuuming process. A steady state is obtained after ~15 h, and the κ of the specimen is evaluated.

The κ of the specimen is determined by matching the experimentally measured temperature profiles with the finite element method (FEM) simulation.[74] Supporting Figure 10 shows the simulation geometry. A specimen is sandwiched between two stainless-steel heat conduits, replicating the experimental setup. Three-dimensional swept-hexahedral meshes are generated, and the experimentally measured temperatures (TC_1 and TC_8) are used as boundary conditions. The interfacial contact resistance between the stainless-steel conduit and specimen ($R_{contact}$) is ignored due to the low κ of the specimen (i.e., $R_{specimen} \gg R_{contact}$). The temperature-dependent κ and emissivity (ε) of stainless-steel heat conduits are experimentally measured in our previous work and used for FEM simulation.[75] The ε of the specimens (GW, VGW, and AGW) is also measured according to the ASTM standard and used for simulation (Supporting Figure 11).[61,76,77]

The κ of the specimen is determined by comparing the experimentally measured temperature profiles and FEM simulation results (Figure 4f).[74] The temperatures of the top heat conduit (TC_1−TC_4), connected to cartridge heaters, show a nonlinear behavior because the radiation heat transfer is non-negligible at high temperatures (~80 °C). In contrast, the temperatures of the bottom heat conduits (TC_5 and TC_8) exhibit an almost linear behavior. The radiation heat transfer is relatively small at low temperatures (~10 °C). The κ of the

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specimen is tuned until the simulation result provides a minimum root-mean-square error (RMSE) in the bottom heat conduit.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(T_{\mathrm{FEM},i} - T_{\mathrm{exp},i})^2} \quad (4)$$

where $T_{\mathrm{FEM},i}$ and $T_{\mathrm{exp},i}$ are the simulated results and experimentally measured temperatures (TC_5–TC_8), respectively. The RMSE values are sufficiently small (≤0.052 °C) for all cases (Supporting Table 3).

Figure 4g shows the κ of GW, VGW, and AGW. The κ of GW is the lowest (3.33 mW m^{-1} K^{-1}). The introduction of additional particles increases κ (VGW = 3.77 mW m^{-1} K^{-1} and AGW = 4.60 mW m^{-1} K^{-1}). This could be partly due to the intrinsically higher κ of particles (e.g., vermiculite = ∼60 mW m^{-1} K^{-1}). The increased contact area due to the addition of particles also increases κ. The κ and κ_{eff} of the heterostructure multilayer specimens (VGW–GW, GW–AGW, and VGW–AGW) are compared in Figure 4h. All the heterostructure specimens provide lower κ, compared with GW, although the κ is increased in VGW and AGW. The κ of VGW-GW is the highest (3.10 mW m^{-1} K^{-1}). The κ_{eff} of VGW-GW is 3.54 mW m^{-1} K^{-1}, and the corresponding κ reduction ratio is 12.4%. Surprisingly, the κ of VGW-AGW is the lowest (2.87 mW m^{-1} K^{-1}) although κ_{eff} is the highest (4.57 mW m^{-1} K^{-1}). The corresponding κ reduction ratio is as high as 37.3%. This could be due to the PDOS mismatch between silica aerogel and vermiculite particles, as discussed in Figure 4b. Note that other factors, such as the spatial distribution of particles and particle–particle contact, could also play a role since particles are impregnated into the porous fibrous network. This needs to be investigated further in the future. This work demonstrates that κ_{solid} can be further decreased by the PDOS mismatch between impregnated dissimilar particles.

CONCLUSIONS

In summary, the κ of bulk-scale heterostructure multilayer composites is reduced by the PDOS mismatch mechanism. The single-layer composites, embedded with PDOS-mismatched AlN, BN, or SiC particles, are alternately stacked to construct heterostructure multilayer composites. The MD simulation reveals that the SR_{SiC}–SR_{BN} combination has the highest PDOS mismatch (i.e., the smallest overlap factor), interfacial temperature drop, and interfacial thermal resistance. The experimental analysis also reveals that the acoustic impedance transmittance is the smallest for the SR_{SiC}–SR_{BN} combination, resulting in the highest κ reduction ratio of 14.9% (stacking number = 8). The independent MD, acoustic impedance, and κ measurement analyses provide close agreement, demonstrating reliability. The PDOS mismatch mechanism is also applied to porous GW for VIP applications. The combination of VGW-AGW, impregnated with PDOS-mismatched particles, provides a high κ reduction ratio of 37.3%, resulting in 2.87 mW m^{-1} K^{-1} in vacuum (total number of layers = 24). This work bridges the gap between nanoscale phonon theory and bulk-scale multilayer composites. It might find applications in refrigerators, household appliances, and building insulation panels.

METHODS

Synthesis of Single-Layer Composites

SR (Wacker Chemie, ELASTOSIL R 401/10, 10.5 g) and the cross-linker (2,5-dimethyl-2,5-di(*tert*-butylperoxy)hexane, KCC Silicon, SH0010U, 1.05 g) are dissolved in THF (Sigma-Aldrich, 186562, 150 mL). AlN (Avention, 24304–00–5, 10 μm, 11.7 g), BN (Sigma-Aldrich, 255475, 1 μm, 8.24 g), or SiC (Sigma-Aldrich, 378097, 74 μm, 11.6 g) particles are then uniformly dispersed in the SR solution by tip sonication (Sonictopia, SONIC 7003, 560 W, 10 min). The corresponding volumetric filler fraction in SR is 25 vol % (Supporting Table 2). The mixture is further stirred using a magnetic bar (40 °C, 500 rpm, 3 h), drop-cast on a Petri dish, and dried overnight to evaporate THF and obtain a putty-like composite. It is then hot-pressed in a Teflon mold (5 MPa, 180 °C, 10 min) and further cured on a hot plate (200 °C, 4 h) to obtain a single-layer composite (SR_{AlN}, SR_{BN}, or SR_{SiC}).

Synthesis of Heterostructure Multilayer Composites

A putty-like composite is obtained using an identical procedure as described above. It is then cold-pressed at 5 MPa to obtain a single-layer composite (thickness = ∼1 mm). The single-layer composites with different fillers are then alternately stacked in three different combinations (SR_{AlN}–SR_{BN}, SR_{SiC}–SR_{AlN}, and SR_{SiC}–SR_{BN}). The total number of stacked layers is changed (2, 4, and 8). The stacked specimen is hot-pressed in a Teflon mold (5 MPa, 180 °C, 10 min) and further cured on a hot plate (200 °C, 4 h) to obtain a multilayer composite.

Impregnation of Glass Wool with Vermiculite or Silica Aerogel Particles

Vermiculite particles (Shin Sung Trading Co., Ltd., average size = 20 μm) are dispersed in deionized water and tip-sonicated for 10 min. The dispersion is then spray-coated onto GW to synthesize VGW. The concentration of vermiculite particles in GW is 16.8 wt %. AGW is commercially obtained (Enersens, skogar ST).

Characterization

The microstructure is characterized by SEM (JEOL, JSM-IT800). The κ of composites is obtained by $\kappa = \alpha\rho C_p$.[60,61] The α is measured by the laser flash method (Netzsch, LFA 467). The ρ value is measured by the Archimedes method (Sartorius, Quinitix224–1SKR). The C_p is measured by differential scanning calorimetry (Netzsch, DSC214). The v is measured by the ultrasonic pulse-echo method (DPR300, longitudinal mode, 10 MHz). The κ of GW, VGW, and AGW are measured using a laboratory-built setup according to the ASTM-D5470 standard.[73,74] Three cartridge heaters, connected to a DC power supply (Toyotech, TEX 200–6, 24.20 V), are used to generate heat flux in the top stainless-steel heat conduit. The bottom stainless-steel heat conduit is employed as a heat sink and connected to a chiller (Daihan Scientific, WCR—P6). Four T-type thermocouples (Omega Engineering, SCP-SS-020-G-2) are inserted into each stainless-steel conduit for the temperature measurement. The thickness and applied pressure on the specimen are measured using thickness (Mitutoyo, 547–315A) and pressure (Curiosity Technology, CMNS-50L) sensors, respectively. The κ is measured in a vacuum, and the chamber pressure is monitored (SUPERBEE, CVM201GCA). The reflected temperature and emissivity of the specimen are calibrated using an infrared camera (FLIR, A325sc) according to the ASTM-E1862 and E1933 standards.[61,76,77]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.6c02949.

Phonon density of state calculation; atomistic models; thermogravimetric analysis; cross-sectional SEM images of multilayer composites; properties of single-layer composites and heterostructure multilayer composites;

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experimentally measured C_p values; interfacial thermal resistance analysis; chamber pressure measurement; FEM simulations; emissivity measurement; unit cell size of composite models in MD simulations; mixture composition of single-layer composites; and root-mean-square error analysis (PDF)

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Author Contributions

‖G.K., J.Y., D.S., and Y.K. contributed equally to this work. G.K., J.Y., D.S., Y.K, J.C., and S.B. conceived and designed the experiments, which were carried out by G.K., J.Y., D.S, M.A., S.H., J.L. Y.K., H.K., and J.C carried out the molecular dynamics simulation. G.K., J.Y., D.S., Y.K, J.C., and S.B. wrote the paper. All authors contributed to data analysis and scientific discussion.

Notes

The authors declare no competing financial interest.

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